

Guidelines and Recommendations to Investigate the Efficacy of a Lower-Limb Prosthetic Device: A Systematic Review

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Abstract—*Introduction* Evaluating the efficacy of a lower-limb prosthetic device (LLPD) is a crucial step in the iterative process of redesigning and product development of new generation prostheses. To date, no standardized experimental methodology for the evaluation of LLPDs is available which complicates the comparison of results. *Objective* This systematic review provides an overview of study designs, tests and outcome measures to evaluate a LLPD. Ultimately, guidelines and recommendations are formulated. *Methods* Two databases were screened on studies evaluating a LLPD, published between 2009 and 2019. *Results* 109 articles were eventually included. We identified 43 outcome measures in four main categories: 53% were biomechanical, 26% physiological, 14% physical performance and 7% psychological. Mainly walking tests were included (78%), of which 31% were performed on a slope, followed by stair climbing (15%) and other functional tasks (15%). *Conclusion* An overview of currently used study designs and experimental protocols is presented, which led to guidelines and recommendations for the evaluation of a LLPD. The main recommendation is to include daily activities in a case-report and in a later stage in a randomized controlled study design. Biomechanical, physiological, physical performance and psychological outcome measures are key for the evaluation of a LLPD.

Index Terms—Evaluation process, lower-limb prosthetics, technology readiness level.

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I. INTRODUCTION

IT IS well-known that lower-limb prosthetic users (LLPUs; individuals with a transfemoral or transtibial amputation) experience reduced functionality during daily life [1]. LLPUs show an altered movement pattern in terms of spatiotemporal, kinematic and kinetic outcome measures [2]. Walking efficiency is negatively affected due to a protective and unstable gait pattern, which encompasses a wider step width and longer stance time of the non-involved side [3]. Therefore, the metabolic cost of LLPUs during walking is higher compared to individuals without an amputation [4]. Consequently, the self-selected walking speed is lower in people with a transfemoral ($1.20 \text{ m}\cdot\text{s}^{-1}$) compared to individuals with a transtibial ($1.35 \text{ m}\cdot\text{s}^{-1}$) and individuals without an amputation ($1.52 \text{ m}\cdot\text{s}^{-1}$) [5]. Several other activities become more challenging as well: taking stairs [6], walking up- or downhill [7], crossing obstacles [8] and taking turns [9]. To reduce aforementioned difficulties, improving prosthetic alignment and the development of novel devices are required [10]. The further optimization of prosthetic devices (greater range of motion, stabilizing and powering joints [11]) would positively influence functional independency and prevent further pathology along the body segments [12].

Preceding a market-proof product, a lower-limb prosthetic device (LLPD) undergoes an iterative process of design, development and construction. In contrast to passive or non-microprocessor-controlled prostheses, quasi-passive or microprocessor-controlled devices improve knee kinematics during stance and swing phases, and they have the ability to adapt to the fluctuations of the walking speed [13]. Recently, technological advancements induce a shift from solely passive or quasi-passive prostheses, to active (actuated joints) and ultimately biomechatronic devices (integration of biology, mechanics and electronics) [14]. Nevertheless, each device aims at restoring the basic lower-limb functions, i.e., stable support, energy storage and propulsion, and shock absorbance [15]–[18].

In the field of robotics, the importance and awareness of benchmarking and standardization of the evaluation process is well-defined [19], e.g., Mudie *et al.* [20] described guidelines for the evaluation and testing of exoskeletons. Additionally, the European Union's 'medical device regulation' (adapted in April, 2017) demands rigorous evaluation of medical devices

TABLE I
NUMBER OF HITS ON KEYWORDS AND COMBINED KEYWORDS IN PUBMED AND WEB OF SCIENCE

Keywords	Hits	
	PubMed	Web of Science
(1) (Amputation OR amputee OR amputees) AND (lower extremity OR lower-limb OR lower-leg) AND (prosthesis/prostheses OR prosthetic design OR assistive device OR robotic prosthesis/prostheses)	1 836	1 812
(2) Physiological measures OR (cognitive AND (response OR function)) OR energy expenditure OR physical exertion OR oxygen consumption OR metabolic cost	402 422	391 982
(3) Biomechanics OR kinematics OR kinetics OR biomechanical measures OR (gait AND (symmetry OR adaptation))	215 422	825 496
(4) Dual task(s) OR dual tasking	4 735	15 775
Combined keywords		
(1) AND (2)	135	154
(1) AND (3)	486	370
(1) AND (4)	48	47
(1) AND ((2) OR (3) OR (4)) *	552	475

*Combined keywords were included in the screening process

before they are considered market-proof [21]. Until now, no unified experimental methodology exists for the clinical evaluation of a LLPD. Current studies determined different biomechanical, physiological and psychological measures during various tasks [22]. Moreover, literature is inconsistent and unclear concerning the implementation of a familiarization trial preceding the evaluation of a LLPD [23].

A model environment to benchmark a LLPD is a championship in which LLPUs compete against each other to complete daily tasks using state-of-the-art technology, e.g., the ‘Cyathlon’ [24]. This competition provides an ideal test situation for different prosthetic systems. However, guidelines and standardized frameworks for the evaluation of LLPDs during the development process are crucial as well. It will eventually help to advance from one ‘technology readiness level’ to the other [25].

Therefore, the aim of the current review was to provide an overview of study designs, experimental tests and outcome measures to evaluate a LLPD in a clinical setting. Ultimately, bottom-up (literature based) guidelines and recommendations on the evaluation process of a LLPD are formulated.

II. METHODS

A. Eligibility Criteria

The review protocol was developed in line with the Preferred Reporting Items for Systematic review and Meta-Analysis Protocols [26]. We applied PICO (population, intervention, comparison and outcome) criteria to include relevant articles. Included studies evaluated unilateral, lower-limb, passive, quasi-passive or active prostheses in humans during daily activities by measuring at least one outcome measure in the categories biomechanics, physiology, psychology or physical performance. Exclusion criteria were papers investigating

participants with a bilateral amputation or an upper-limb prosthesis. Individuals with a Medicare Functional Classification K-level 0 or 1 were not included either [27]. Only original studies written in English were considered.

B. Search Strategy and Data Collection Process

A comparable systematic review of Sagawa *et al.* [22] already reported the relevant outcome measures between 1979 and May 2009. Therefore, our systematic search strategy (Table I) started from May 2009 until May 2019. Two electronic databases, Pubmed and Web of Science, were searched to collect articles published in the last decade.

The first screening process for in- and exclusion of articles was performed with application of PICO criteria to the title and abstract. The remaining articles were possible eligible and screened on full text with the same criteria. This data collection process is presented in Fig. 1 [28].

C. Quality Assessment

The methodological quality of the included articles was assessed according to their study design. The following scoring lists were applied: ‘the national institute for health and clinical excellence’ (case report) [29]; ‘strengthening the reporting of observational studies in epidemiology’ (crossover and cross-sectional) [30] and ‘the quality assessment of diagnostic accuracy studies’ (diagnostic) [31]. Scoring lists were made dichotomous. If one question consisted of multiple parts, all questions were merged and scored as one. This means the answer was ‘yes’ if the answers to all the questions were ‘yes’ and ‘no’ if one or more questions were answered ‘no’. Studies were independently assessed by three reviewers (BT, JG and SDB) and disagreements were solved by consensus. A summary score and a mean percentage for each article and study design were calculated. A score of >75% indicated ‘strong’

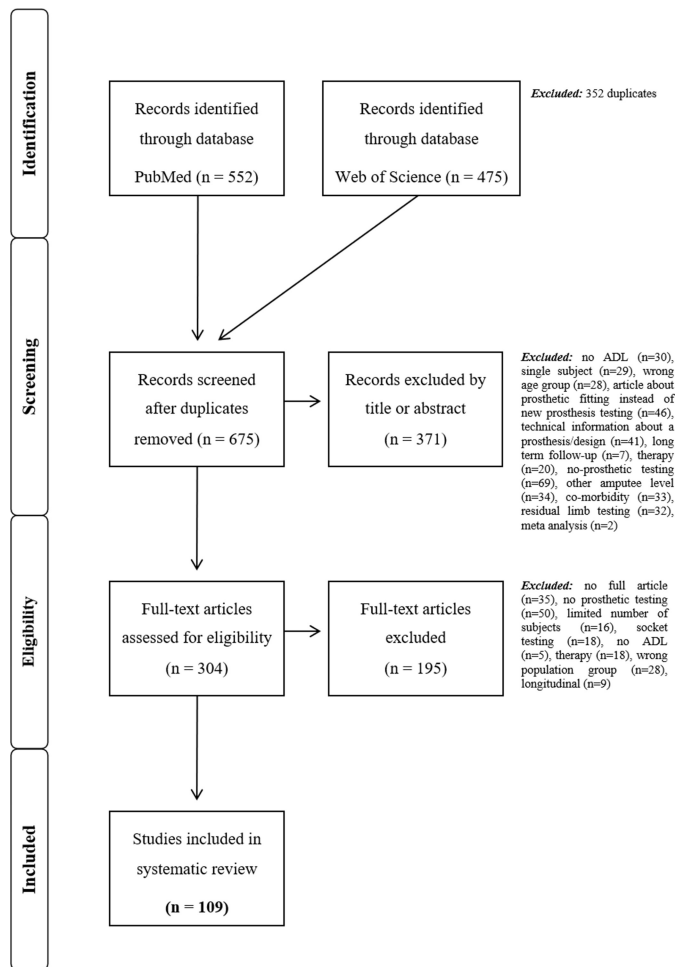


Fig. 1. Selection process for the inclusion of research articles in this systematic review.

quality, a score of 55–75% indicated ‘moderate’ quality, and a score <55% indicated ‘low’ quality.

III. RESULTS

A. Study Selection and Data Quality

The selection process is presented in Fig. 1. A total of 1027 articles were identified, based on our predetermined search method and 675 articles remained after removal of duplicates, publication year and language. These articles were further screened on in- and exclusion criteria, primarily on title and abstract and secondly on full text. A total of 109 eligible articles were eventually included. Quality assessment of these articles determined that 61 articles were of ‘strong’ and 48 articles of ‘moderate’ quality.

B. Study Design

In general, the included studies applied a crossover (56%) [32]–[91], case report (25%) [7], [23], [43], [92]–[112], cross-sectional (14%) [113]–[126], cohort (4%) [127]–[130], diagnostic (1%) [131] or a randomized controlled (1%) [132] study design. A control group (individuals without an amputation) [7], [33], [35], [41], [44], [47], [49], [67], [76], [80],

[87], [93]–[95], [98], [100], [101], [106], [107], [109], [111], [114], [118], [126], [132], [133] was implemented in 23% of the included articles, while 28% counterbalanced the different conditions [32]–[35], [40], [42], [46], [47], [51], [56], [57], [59], [61], [63], [64], [69]–[72], [76], [78], [79], [81], [83], [84], [96], [100], [113], [132], [134]. A familiarization trial (= one day) [34]–[37], [39]–[41], [43], [45], [46], [50], [53], [54], [68], [69], [72], [77], [81], [82], [85], [88], [89], [91], [92], [100], [101], [104], [107], [111]–[113], [115], [117], [119], [121]–[124], [129], [131], [133] was included in 38% of the included articles. Other studies reported a familiarization period of more than one day (39%) [7], [39], [45], [54], [62]–[64], [66], [67], [69], [70], [72], [75], [77], [79], [82], [85], [86], [90], [94], [96], [101], [109], [114], [116], [121], [126], [130], [132]–[135] or no familiarization (24%) [23], [32]–[34], [40], [42], [48], [49], [51]–[53], [58], [61], [65], [68], [71], [76], [78], [82], [92], [95], [102]–[107], [110], [112], [122], [124], [127], [128], [136], [137]. An overview of the enclosed articles, including their study design, objective, population, tested device, intervention and outcome measures is outlined in Table II.

C. Experimental Tests

The included articles mostly executed walking trials (78%). Treadmill (37%, n=40) [32], [34], [37], [38], [40], [41], [46], [47], [52], [53], [56], [57], [60], [64], [71], [73], [89], [92]–[95], [97], [98], [100]–[102], [107], [108], [110], [113], [121], [124], [125], [127], [129]–[131], [134], [138] and hall-way walking (42%, n=45) [23], [36], [37], [40], [42], [43], [48], [50]–[55], [57]–[60], [64], [65], [68], [74], [75], [77], [80], [82]–[84], [87], [89], [90], [95], [98], [99], [104], [111], [115], [116], [118], [119], [122], [123], [125], [126], [128], [135] with a variation in length of the hallway (mean: 14 ± 8 m, range: 1–55 m) were assessed. Walking tests were performed at a self-selected [23], [32], [35], [36]–[97], [98], [99], [101]–[103], [109], [111], [119], [121], [124], [127], [129], [131], [133], [134] or at a fixed walking speed (range: 0.50 – 1.75 m.s⁻¹) [34], [40], [41], [43], [57], [85], [89], [92], [101], [104], [107], [109], [113], [122]–[124], [128], [130], [138]. Some walking trials lasted two [37], [46], [55], [131], six [40], [42], [43], [64], [66], [86], [119] or more than six minutes [56], [94], [104], [121], [123] while the other walking trials did not mention the total duration. Slope ascending and descending trials were conducted in 31% (n=34) of the studies [7], [32], [38], [40], [42], [49], [52], [57], [60], [63], [65], [68]–[70], [73], [75], [78], [79], [81], [87]–[91], [94], [99], [102], [106], [108], [115], [121], [127], [129], [137] including 5.96 ± 2.13 m long ramps with a mean ascending and descending inclination of $10 \pm 3\%$. The stair climbing test was performed in 15% (n=16) of the included articles [33], [34], [45], [48], [58], [59], [77], [80], [82], [85], [86], [94], [95], [110], [124], [132]. An average staircase of 8-steps with a mean rise of 27.7 ± 1.2 cm and run of 18.3 ± 0.3 cm was used. Finally, 15% (n=16) of the selected articles [35], [39], [45], [47], [78], [80]–[82], [97], [114], [133] executed more functional tests, such as the timed up & go

TABLE II
OVERVIEW AND SUMMARY OF THE STUDY DESIGN, OBJECTIVE(S), POPULATION CHARACTERISTICS, TESTED DEVICE(S), INTERVENTION(S) AND OUTCOME MEASURES OF INCLUDED ARTICLES*

Author	Study design	Objective(s)	Population (n)	Tested device(s)	Intervention(s)	Outcome measures
Abouhossein <i>et al.</i> (2019)	Case report	To investigate the short-term process of adaptability for two novel prostheses	TFA(1) NA(4)	NMPK MPK	Hallway walking	Ground reaction forces, range of motion, speed
Adamczyk <i>et al.</i> (2017)	Cross-sectional	To determine whether forefoot or hindfoot stiffness plays a role in biomechanical adaptations	TTA(6)	MPA	Treadmill walking	Centre of motion, ground reaction forces, power, range of motion, torque
Agrawal <i>et al.</i> (2015)	Crossover	To determine differences of four prosthetic devices during slope walking	TTA(10)	MPK MPA	Slope walking	Centre of motion, ground reaction forces, speed
Aldridge <i>et al.</i> (2012)	Crossover	To determine if providing extra power in the ankle joint normalized biomechanical outcome measures	TTA(11) NA(11)	MPA PA	Stair climbing	Cadence, power, range of motion, speed, stance time, swing time, torque
Aldridge <i>et al.</i> (2014)	Crossover	To compare stair ascent strategies between two prostheses	TFA(14) NA(10)	MPK	Stair climbing	Cadence, ground reaction forces, power, range of motion, stair assessment index, torque
Andrysek <i>et al.</i> (2011)	Crossover	To develop and evaluate a new type of stance phase controlled prosthetic knee joint	TFA(14)	MPK	Hallway walking	Heart rate, speed
Andrysek <i>et al.</i> (2016)	Crossover	To evaluate the automatic stance phase lock knee mechanism	TFA(10)	MPK	Hallway walking	Heart rate, speed, questionnaires
Bai <i>et al.</i> (2017)	Crossover	To evaluate the function of a novel ankle device on level and camber surface walking	TFA(5) NA(12)	NMPA MPA	Hallway walking	Questionnaires, range of motion, speed, stance time, step length, stride length
Bai <i>et al.</i> (2018)	Crossover	To investigate the changes in biomechanics of a novel ankle during slope walking	TTA(5) NA(14)	MPA	Slope walking	Power, questionnaires, range of motion, speed, torque
Barr <i>et al.</i> (2012)	Case report	To document biomechanical changes over a seven-years period with three different prostheses	TFA(1)	MPK	Hallway walking	Double stance time, ground reaction forces, speed, step length
Bell <i>et al.</i> (2016)	Cohort	To compare different prostheses during slope walking	TFA(21)	MPK	Slope walking	Centre of motion, ground reaction forces, power, range of motion, step length
Bellman <i>et al.</i> (2012)	Crossover	To investigate the immediate biomechanical effects after transition to a novel device	TFA(11)	MPK	Balance task, Hallway, Treadmill and Slope walking, Stair climbing	Centre of pressure, ground reaction forces, power, range of motion, speed, stance time, step length, swing time, torque
Bellman <i>et al.</i> (2018)	Crossover	To investigate the functional performance of the current standard of two novel knee joints during daily life	TFA(6)	MPK	Hallway walking Slope walking Stair climbing Treadmill walking	Power, ground reaction forces, range of motion, speed, stance time, swing time, torque
Brandt <i>et al.</i> (2017)	Crossover	To investigate the influence of different powered knee mechanics during walking with and without additional load	TFA(5)	PK		Centre of motion, ground reaction forces, power, range of motion, rating of perceived exertion, stance time, stride length, swing time, torque
Burnfield <i>et al.</i> (2012)	Crossover	To assess performance, function and balance differences during slope walking between two prostheses	TFA(10)	MPK	Slope walking Timed up & go	Cadence, duration, electromyography, range of motion, speed, stance time, stride length, questionnaires
Creyllman <i>et al.</i> (2016)	Crossover	To assess differences in gait pattern between two prostheses	TFA(4)	PK	Hallway walking	Ground reaction forces, range of motion, speed, stance time, questionnaires, visual analogue scale comfort
Culver <i>et al.</i> (2018)	Case report	To develop a control system for a powered prosthesis during stair climbing	TTA(1)	PA	Stair climbing	Power, range of motion, torque

TABLE II
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Author	Study design	Objective(s)	Population (n)	Tested device(s)	Intervention(s)	Outcome measures
De Asha <i>et al.</i> (2014)	Crossover	To determine the difference in centre of mass braking effect between two prostheses	TFA(8) TTA(11)	NMPA NMPK	Hallway walking	Centre of motion, centre of pressure, power, speed, stance time, step length, swing time, swing time, torque Distance, heart rate, ventilatory exchange, VO_2 , VCO_2 , questionnaires
De Pauw <i>et al.</i> (2018)	Crossover	To evaluate a novel ankle prosthesis during walking at different speeds	TFA(6) TTA(6) NA(6)	MPA	Treadmill walking	Distance, heart rate, hill assessment index, stair assessment index, respiratory exchange ratio, VO_2 , VCO_2 , questionnaires
Delussu <i>et al.</i> (2013)	Crossover	To determine the effectiveness of two assistive devices	TTA(10)	PA	Slope walking Treadmill walking	Distance, heart rate, hill assessment index, stair assessment index, respiratory exchange ratio, VO_2 , VCO_2 , questionnaires
Eberly <i>et al.</i> (2014)	Crossover	To compare hallway walking between two prostheses	TFA(10)	NMPK MPK	Hallway walking	Cadence, electromyography, ground reaction forces, power, range of motion, speed, stance time, stride length, swing time, torque Duration, power, questionnaires, speed, range of motion, torque, VO_2
Esposito <i>et al.</i> (2018)	Case report	To compare a novel powered prosthesis to current passive devices	TTA(3)	MPA	Balance task Hallway walking	Centre of pressure, double stance time, ground reaction forces, metabolic cost, power, stance time, torque, VO_2 , VCO_2
Feng <i>et al.</i> (2017)	Crossover	To determine gait symmetry and metabolic cost during terrain ambulation with a redesigned active prosthesis	TTA(3)	PA	Slope walking Treadmill walking	Distance, power, range of motion, speed, stance time, step length, swing time, torque, VO_2 , VCO_2 , questionnaires
Ferris <i>et al.</i> (2012)	Case report	To evaluate gait mechanics, physical performance and user satisfaction with a novel ankle prosthesis	TTA(11) NA(11)	PA	Treadmill walking	Electromyography, ground reaction forces, range of motion
Fey <i>et al.</i> (2011)	Crossover	To determine the influence of an ESAR foot stiffness on gait characteristics	TTA(12)	MPA	Treadmill walking	Ground reaction forces, power, range of motion, speed, torque
Fradet <i>et al.</i> (2010)	Case report	To evaluate the biomechanical effects of adaptation of the prosthetic ankle during ramp ambulation	TTA(16) NA(16)	PA	Slope walking	Distance, speed, questionnaires
Gailey <i>et al.</i> (2012)	Crossover	To determine whether self-reported questionnaires could detect functional differences between four prostheses	TTA(10)	MPK	Hallway walking	Distance, heart rate, rating of perceived exertion, respiratory exchange ratio, step frequency, step length, ventilatory exchange, VO_2 Ground reaction forces
Gjovaag <i>et al.</i> (2013)	Case report	To compare physiological outcomes during walking	TFA(12) NA(10)	MPK	Treadmill walking	Ground reaction forces
Grabowski <i>et al.</i> (2013)	Crossover	To compare kinetic outcome measures between two prosthetic devices during different walking speeds	TTA(7) NA(7)	PA	Hallway walking	Ground reaction forces, power, range of motion, stance time, step length, stride time, torque
Grimmer <i>et al.</i> (2017)	Case report	To compare two prosthetic devices during walking regarding biomechanical outcome measures	TTA(1)	PA	Treadmill walking	Centre of pressure, duration, ground reaction forces, range of motion, questionnaires
Hansen <i>et al.</i> (2014)	Crossover	To examine the effect of a prosthetic foot rocker on balance and mobility	TTA(12)	NMPA	Balance task, Hallway walking, L-test, Sit to stand	Ground reaction forces, questionnaires, range of motion, speed, swing time
Hasenoehtl <i>et al.</i> (2017)	Crossover	To assess safety, function and quality of life in low-active ambulators when switching from prosthesis	TFA(18)	MPK	Balance task, Hallway and treadmill walking, Timed up & go	Centre of motion, duration, ground reaction forces, power, range of motion, torque
Highsmith <i>et al.</i> (2011)	Cross-sectional	To evaluate kinetic asymmetry during sit to stand tasks	TFA(21) NA(7)	MPK PK	Sit to stand	Questionnaires
Highsmith <i>et al.</i> (2016a)	Crossover	To determine if laboratory determined benefits of a P-MPK are detectable with clinical outcome assessments	TFA(20)	MPK	Balance task Hallway walking Stair climbing	Duration, questionnaires, rating of perceived exertion
Highsmith <i>et al.</i> (2016b)	Randomized controlled trial	To determine performance and preferences of three prosthetic feet during an obstacle course	TTA(14) NA(14)	NMPA	Obstacle course	

TABLE II
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Author	Study design	Objective(s)	Population (n)	Tested device(s)	Intervention(s)	Outcome measures
Highsmith <i>et al.</i> (2017)	Crossover	To determine if use a MPK would reduce the degree of asymmetry of peak stance knee flexion moment	TFA(20)	MPK	Slope walking	Ground reaction forces, range of motion, speed, stance time
Hood <i>et al.</i> (2018)	Case report	To investigate the effects of a powered knee joint on gait kinematics	TFA(1) NA(1)	PK	Treadmill walking	Heart rate, power, step length, stride length, torque
Houdijk <i>et al.</i> (2018)	Cross-sectional	To investigate if ESAR feet improve biomechanical outcomes during walking	TTA(15)	NMPA	Hallway walking	Centre of motion, ground reaction forces, power, range of motion, speed, step length, torque
Howard <i>et al.</i> (2017)	Case report	To examine changes in sway under increasing postural challenge between single and dual task conditions	TTA(13) NA(15)	MPA	Balance task	Centre of pressure, ground reaction forces, step length
Huang <i>et al.</i> (2016)	Cross-sectional	To determine whether an amputee would adapt their residual muscle activation patterns with an active prosthesis	TTA(5)	PA	Treadmill walking	Electromyography, power, range of motion, torque
Ingraham <i>et al.</i> (2016)	Case report	To investigate how powered ankle and knee joints contribute to walking mechanics of amputees	TTA(3)	PA PK	Hallway walking	Ground reaction forces, range of motion, stance time, swing time, speed, step width, stride length
Inoue <i>et al.</i> (2013)	Cross-sectional	To investigate the effects of the inertial properties of a prosthesis during stair ascending	TFA(5)	MPK	Stair climbing	Power, torque
Johnson <i>et al.</i> (2014)	Crossover	To investigate whether a hydraulically ankle would increase toe clearance	TTA(21)	MPA	Treadmill walking	Range of motion, speed, stance time, swing time
Kaufman <i>et al.</i> (2012)	Crossover	To compare the gait symmetry between NMPK and MPK joints	TFA(15)	MPK	Hallway walking	Range of motion, speed, stance time, swing time
Kark <i>et al.</i> (2011)	Cross-sectional	To compare suitability of four gait summaries in people with a lower-limb amputation	TFA(8) TTA(12)	MPA	Hallway walking	Ground reaction forces, range of motion, questionnaires
Kendell <i>et al.</i> (2016)	Crossover	To examine gait adaptations across various walking conditions	TFA(11)	MPK	Treadmill walking	Centre of pressure, double stance time, ground reaction forces, single stance time
Klodd <i>et al.</i> (2010)	Crossover	To compare five types of flexible prostheses	TTA(13)	MPA	Treadmill walking	Oxygen cost, speed, questionnaires
Knaepen <i>et al.</i> (2015)	Crossover	To assess cognitive workload during walking	TFA(24)	MPK	Treadmill walking	Breathing frequency, cadence, heart rate, ground reaction forces, stance time, swing time, questionnaires
Koehler-McN <i>et al.</i> (2018)	Crossover	To determine if the bimodal ankle-foot system can improve various aspects of standing balance and mobility	TTA(18)	PA	Balance task Hallway walking L-test	Centre of motion, questionnaires, speed
Lacruz <i>et al.</i> (2016)	Crossover	To compare patient satisfaction and energy expenditure during gait between two prosthetic feet	TTA(15)	MPA	Treadmill walking	Questionnaires, VO ₂ , VCO ₂
Lansade <i>et al.</i> (2018)	Crossover	To evaluate the effect of a MPK joint on dynamic balance	TFA(27)	MPK	Timed up & go	Duration, questionnaires
Ledoux <i>et al.</i> (2016)	Crossover	To assess the differences in stair ascending between a passive and active device	TFA(3)	PK	Hallway walking Stair climbing	Duration, power, questionnaires, range of motion, torque, VO ₂ , VCO ₂
Lura <i>et al.</i> (2015)	Crossover	To determine biomechanical differences between two prostheses	TFA(20) NA(5)	MPK	Slope walking Treadmill walking	Ground reaction forces, range of motion, speed, stance time, swing time

TABLE II
CONTINUED

Author	Study design	Objective(s)	Population (n)	Tested device(s)	Intervention(s)	Outcome measures
Lura <i>et al.</i> (2017)	Crossover	To determine the difference between two prostheses during stair climbing	TFA(20)	MPK	Stair climbing	Ground reaction forces, power, range of motion, torque
Määret <i>et al.</i> (2010)	Crossover	To evaluate the influence of the type of knee during gait	TFA (29) NA (15)	MPK	Hallway walking	Cadence, ground reaction forces, range of motion, speed, stance time, stride length, swing time
Major <i>et al.</i> (2016)	Crossover	To observe the effects of stiffness on gait parameters during walking	TTA(5)	NMPA	Hallway walking Slope walking	Range of motion, speed, step width, stance time, swing time, questionnaires
Martinez <i>et al.</i> (2011)	Case report	To evaluate the metabolic cost while walking with an active prosthesis	TFA(1)	PK	Hallway walking Treadmill walking	Speed, VO ₂
McDonald <i>et al.</i> (2018)	Crossover	To compare energy expenditure at three different speeds wearing two novel prosthetic feet	TTA(27)	NMPA	Treadmill walking	Speed, VO ₂
McGrath <i>et al.</i> (2018)	Case report	To compare two prosthetic feet during slope walking	TTA(4)	NMPA MKA	Slope walking	Power, speed, stance time, torque
Meier <i>et al.</i> (2012)	Crossover	To evaluate the manoeuvrability and efficiency of three different prostheses	TFA(12)	MPK	Obstacle course	Duration, heart rate
Mengelkoch <i>et al.</i> (2016)	Case report	To determine energy costs and ambulatory performance differences during walking and running with an ESAR foot	TFA(3) NA(3)	MPA	Treadmill walking	Speed, VO ₂ , VCO ₂
Mishra <i>et al.</i> (2019)	Crossover	To quantify the gait performance of a novel prosthetic device	TFA(11)	MPK	Hallway walking	Range of motion, stance time, swing time
Molei <i>et al.</i> (2017)	Case report	To demonstrate the effectiveness of a sit to stand controller	TFA(1)	PK	Sit to stand	Power, range of motion, torque
Möller <i>et al.</i> (2017)	Cross-sectional	To measure self-efficacy and the relation between outcome measures with two different prostheses	TFA(42)	MPK	Hallway walking	Questionnaires
Montgomery <i>et al.</i> (2018)	Cross-sectional	To compare passive and powered prostheses for energy consumption and step-to-step transition on slope walking	TTA(10)	MPA	Slope walking	Centre of motion, ground reaction forces, power, torque, VO ₂ , VCO ₂
Morgan <i>et al.</i> (2015)	Cross-sectional	To quantify changes in walking associated with the addition of a concurrent cognitive task with a novel device	TFA(14) NA(14)	MPK	Hallway walking	Cadence, questionnaires, range of motion, reaction time, speed, step length, step width, stride length
Mulder <i>et al.</i> (2013)	Case report	To test a prosthetic device for Syme or Pirogoff amputees with an overall installation of 2 cm	TTA(6)	NMPA	Treadmill walking Slope walking	Cadence, double stance time, ground reaction forces, questionnaires, stance time, step length, speed
Nickel <i>et al.</i> (2014)	Case report	To assess the ability of adapting a prosthetic foot while walking on a slope	TTA(1)	MPA	Slope walking	Power, questionnaires, range of motion, speed, torque
Pauley <i>et al.</i> (2011)	Case report	To determine the influence of a cognitive task on motor performance	TTA(10) NA(13)	MPA	Treadmill walking	Reaction time
Pew <i>et al.</i> (2016)	Crossover	To determine if a novel prosthesis can reduce loading on the residual limb during walking	TTA(5)	MPA	Hallway walking L-test	Ground reaction forces, power, speed, torque
Pickle <i>et al.</i> (2014)	Case report	To evaluate dynamic balance during stair climbing	TTA(9) NA(9)	MPA	Stair climbing	Ground reaction forces, power, range of motion, torque

TABLE II
CONTINUED

Author	Study design	Objective(s)	Population (n)	Tested device(s)	Intervention(s)	Outcome measures
Pickle <i>et al.</i> (2016)	Case report	To investigate the difference in whole body angular momentum between a quasi-passive versus active prosthesis	TTA(10) NA(10)	MPA	Slope walking	Ground reaction forces, power, range of motion, speed, torque
Portnoy <i>et al.</i> (2012)	Crossover	To compare the internal stresses in the residuum of amputees between their current and novel prosthetic foot	TTA(10)	NPA	Balance task, Hallway and slope walking, stair climbing	Cadence, ground reaction forces
Prinsen <i>et al.</i> (2017a)	Crossover	To compare walking with passive and quasi-passive devices	TFA(9)	MPK	Treadmill walking	Centre of motion, double stance time, range of motion, speed, step length
Prinsen <i>et al.</i> (2017b)	Crossover	To investigate the added value of a novel device in planned gait termination	TFA(10)	MPK	Hallway walking	Centre of motion, centre of pressure, double stance time, power, range of motion, single stance time, torque
Quesada <i>et al.</i> (2016)	Cohort	To determine the relationship between prosthetic ankle push-off work and metabolic energy during walking	TTA(6)	PA	Treadmill walking	Centre of motion, ground reaction forces, power, questionnaires, range of motion, speed, torque, VO ₂ , VCO ₂
Raschka <i>et al.</i> (2015)	Crossover	To determine if the reported preference was related to a biomechanical characteristic of the foot and if it resulted in increased activity	TTA(11)	MPA	Hallway walking	Cadence, power, questionnaires, stance time, torque
Ray <i>et al.</i> (2018)	Case report	To examine changes in lower-limb mechanical work during walking following a three-week adaptation to a new prosthesis	TTA(22)	MPA	Hallway walking	Ground reaction forces, stride length, swing time
Rosenblatt <i>et al.</i> (2014)	Crossover	To determine the contribution of an active dorsiflexing prosthesis	TTA(8)	PA	Treadmill walking Slope walking	Ground reaction forces, range of motion, speed
Rock <i>et al.</i> (2018)	Crossover	To determine if changes in push-off mechanics affect stride-to-stride fluctuations	TTA(22)	NMPA	Hallway walking Treadmill walking	Power, questionnaires, speed, torque
Roerdink <i>et al.</i> (2014)	Diagnostic	To investigate the effectiveness of a new measurement tool 'gaitography'	TFA(6) TTA(3)	MPA	Treadmill walking	Centre of pressure, distance, double stance time, ground reaction forces, step length, speed, single stance time, stance time, step width
Schmalz <i>et al.</i> (2018)	Crossover	To investigate biomechanical parameters when walking on a ramp that change abruptly	TTA(4) NA(10)	MPA	Slope walking	Power, range of motion, speed, stance time, torque
Segal <i>et al.</i> (2012)	Cohort	To evaluate the effect of an ESAR foot on amputee gait	TTA(7)	MPA	Hallway walking	Centre of motion, ground reaction forces, range of motion, step length, speed, VO ₂
Shepherd <i>et al.</i> (2017)	Case report	To investigate biomechanical outcome measures comparing different stiffness of a prosthetic foot.	TTA(1)	MPA	Slope walking Treadmill walking	Power, range of motion, speed, stance time, swing time, torque
Shultz <i>et al.</i> (2017)	Case report	To describe walking parameters with a novel powered prosthesis during various slopes	TTA(1)	PA	Slope walking	Power, ground reaction forces, range of motion, speed, torque
Shultz <i>et al.</i> (2018)	Crossover	To describe to which extent the controller of a prosthesis affects walking on uneven terrain	TTA(5)	PA	Balance task	Power, ground reaction forces, range of motion, torque
Simon <i>et al.</i> (2013)	Crossover	To develop and evaluate a powered knee during multiple ambulation modes	TFA(3)	PK	Hallway walking Slope walking Stair climbing	Power, range of motion, stance time, swing time, torque
Simon <i>et al.</i> (2014)	Crossover	To assess biomechanical differences between two prostheses during daily activities	TFA(5) TTA(1)	PA PK	Hallway walking Slope walking Stair climbing	Power, range of motion, stance time, torque

TABLE II
CONTINUED

Author	Study design	Objective(s)	Population (n)	Tested device(s)	Intervention(s)	Outcome measures
Siarholm <i>et al.</i> (2010)	Cross-sectional	To compare oxygen consumption between prostheses during walking on different inclinations	TFA(8)	MPK	Treadmill walking	Duration, heart rate, rating of perceived exertion, VO ₂ , VCO ₂
Struchkov <i>et al.</i> (2015)	Cohort	To compare two prosthetic feet during ramp descent	TTA(9)	MPA	Slope walking	Centre of pressure, double stance time, ground reaction forces, range of motion, single stance time
Svoboda <i>et al.</i> (2012)	Crossover	To determine the influence of a prosthetic foot on ground reaction forces variability during walking	TTA(11)	MPA	Hallway walking	Ground reaction forces, stance time
Takahashi <i>et al.</i> (2015)	Case report	To compare the mechanical energy profiles of two prostheses	TTA(1) NA(11)	PK	Treadmill walking	Centre of motion, centre of pressure, power, range of motion, speed, torque
Thiele <i>et al.</i> (2014)	Crossover	To evaluate the quality of stance and swing phase between four prosthetic knees	TTA(3)	MPK	Hallway walking	Ground reaction forces, range of motion, speed, stance time, swing time
Thiele <i>et al.</i> (2019)	Crossover	To show different control mechanism of a microprocessor knee joint	TFA(4)	MPK	Hallway walking	Range of motion, stance time, speed
Turcot <i>et al.</i> (2013)	Crossover	To compare two prosthetic feet during walking	TTA(15)	MPA	Hallway walking	Ground reaction forces, power, range of motion, step frequency, torque
Ventura <i>et al.</i> (2010)	Crossover	To compare the effect of adding the ESAR ankle on the muscle activity	TTA(12)	MPA	Hallway walking	Electromyography, ground reaction forces, power, range of motion, torque
Wezenberg <i>et al.</i> (2014)	Cross-sectional	To determine whether ESAR feet are able to reduce mechanical energy during step-to-step transition	TTA(15)	MPA	Hallway walking	Centre of motion, ground reaction forces, power, range of motion, step length, speed, stride time, stance time, torque
Williams <i>et al.</i> (2016)	Cross-sectional	To determine the impact on gait biomechanics with an active variable impedance prosthetic knee	TFA(5)	MPK	Hallway walking	Ground reaction forces, power, range of motion, step length, torque
Wolf <i>et al.</i> (2012)	Case report	To compare biomechanics during stair climbing and slope walking between two prostheses	TFA(5) NA(5)	MPK PK	Slope walking Stair climbing	Ground reaction forces, power, range of motion, step length
Wujiang <i>et al.</i> (2018a)	Cross-sectional	To quantitatively compare the oxygen consumption in amputees wearing intelligent prosthetic knees	TFA(6)	MPK	Treadmill walking	Speed, stance time
Wujiang <i>et al.</i> (2018b)	Cross-sectional	To analyse the plantar pressure with a novel device	TFA(10)	MPK	Treadmill walking	Speed, VO ₂
Wurdeman <i>et al.</i> (2017)	Crossover	To determine changes due to use of low-activity and high-activity ESAR feet	TTA(24)	NMPA	Hallway walking	Centre of pressure, ground reaction forces, stance time, speed
Yusuke <i>et al.</i> (2018)	Crossover	To clarify the biomechanical effect of a novel prosthesis during slope walking	TFA(7)	MPK	Slope walking	Distance
Zelik <i>et al.</i> (2011)	Case report	To determine the effect of prosthetic foot mechanics on gait pattern	TTA(5) NA(5)	MPA	Treadmill walking	Ground reaction forces, power, range of motion, stance time, speed, stride length, swing time, torque
			Mean sample size: 10			Centre of motion, ground reaction forces, power, range of motion, VO ₂ , VCO ₂

*Abbreviations: **MPA** microprocessor-controlled ankle, **MPK** microprocessor-controlled knee, **ESAR** energy storing and return, **NA** individuals without an amputation, **NMPA** non-microprocessor-controlled ankle, **NMPK** non-microprocessor-controlled knee, **PA** powered ankle, **PK** powered knee, **SSWS** self-selected walking speed, **TFA** individuals with a transfemoral amputation, **TTA** individuals with a transtibial amputee, **VO₂** oxygen consumption, **VCO₂** carbon dioxide production.

test [38], [46], [58], [91], the sit to stand test [112], [114], the L-test [65], [68], and balance [46], [52], [65], [69], [76], [89], [97], [132], [135] and dual tasks [54], [65], [126], [134]. Dual tasks involved backward counting or subtracting seven from a three-digit number during walking.

D. Outcome Measures

Forty-three outcome measures were extracted from the included articles, from which 53% (n=23) were biomechanical, 26% (n=11) physiological, 14% (n=6) physical performance and 7% (n=3) psychological (Fig. 2). Frequency analyses were described below per category (biomechanical, physical performance, physiological and psychological). Both the percentages (%) and absolute values (n) were presented.

1) *Biomechanical Measures*: In general, 83% (n=90) of the articles gathered biomechanical measures categorized in kinetic (89%, n=89) [7], [14], [23], [32]–[36], [39], [41], [44], [46], [47], [52], [54], [57], [59]–[63], [65], [68]–[74], [76], [78]–[80], [82], [83], [85], [87]–[90], [92], [94]–[99], [101]–[103], [105]–[118], [122], [123], [125], [127], [128]–[131], [133], [135]–[139] kinematic (77%, n=69) [7], [14], [23], [33]–[36], [39]–[41], [44]–[47], [49]–[51], [57], [59]–[63], [67], [70], [71], [73], [76]–[79], [81]–[83], [85], [87]–[92], [94]–[97], [101], [103], [105]–[110], [112]–[115], [117], [118], [122], [123], [126], [127], [129], [130], [136]–[139] and spatiotemporal data (60%, n=54) [33]–[35], [37], [40], [44], [47], [49]–[54], [57]–[60], [63], [64], [68]–[70], [73], [74], [76], [77], [80], [83], [84], [88], [92], [94], [95], [100], [101], [103], [105]–[107], [109], [115], [118], [119], [121], [122], [125], [127]–[130], [131], [135], [138]. Main spatiotemporal data were collected with a force platform: stance time (65%, n=35), step length (41%, n=22) and cadence (24%, n=13). Other spatiotemporal measures, such as step frequency and width, stride length, swing and single or double stance time were gathered in less than 25% of the studies. Kinematic measures were conducted with motion capture systems: joint angles at different times of the gait cycle; range of motion of the ankle, knee and hip (100%, n=69), trunk (13%, n=9) and upper-limb (10%, n=7). Centre of mass data was assessed in 20% (n=14) of the studies to calculate the angular momentum. The most frequently examined kinetic measures were ground reaction forces (74%, n=59, centre of pressure (15%, n=12), joint torque and power of the ankle (between the shank and the foot), knee (between femur and tibia) and hip (between trunk and femur) (59%, n=47 and 61%, n=49; respectively).

2) *Physical Performance Measures*: Physical performance measures [32]–[40], [44], [46], [48], [49], [54], [64]–[67], [69]–[71], [73], [75], [76], [79], [81]–[84], [87], [90], [92]–[97], [99], [101]–[103], [105]–[109], [111], [114], [115], [117], [120]–[122], [125], [127], [128], [130], [131], [133], [134], [136], [138] were reported in 67% (n=73) of the included articles. The most common measures were speed (78%, n=57) and duration (12%, n=9), followed by distance (10%, n=7), the hill and stair assessment index (3%, n=2) and reaction time (3%, n=2).

3) *Physiological Measures*: Physiological measures were gathered in 24% (n=26) of the included articles. The most cited respiratory measure (73%, n=19) [37], [38], [40], [43], [53], [54], [56], [59], [64], [72], [93], [96], [100], [101], [104], [115], [121], [124], [130] was oxygen consumption (95%, n=18) to indirectly determine energy expenditure, followed by carbon dioxide production (58%, n=11), oxygen cost (26%, n=7), respiratory exchange ratio (16%, n=4), metabolic rate (16%, n=4), breathing frequency (16%, n=3) and ventilatory equivalent (17%, n=3). The only cardiovascular measure (40%, n=10) was heart rate (100%, n=10) [37], [38], [54]–[56], [65], [66], [94], [99], [121]. Some articles also gathered electrophysiological measures (28%, n=7) [39], [41], [54], [85], [91], [130], [138], i.e., electromyography (EMG, 86%, n=6), skin conductance (14%, n=1) and electroencephalography (14%, n=1).

4) *Psychological Measures*: Psychological and cognitive measures were investigated in 28% of the included articles (n=30) [35], [36], [38], [42], [46]–[48], [50], [56], [57], [60], [62], [65], [66], [71], [83], [87], [88], [90], [96], [101], [103], [104], [109], [116], [117], [122], [130], [132], [134], i.e., the rating of perceived exertion scale (RPE, 17%, n=5), the visual analogue scale for comfort and fatigue (17%, n=5) and questionnaires (87%, n=26).

IV. DISCUSSION

The iterative process of design, construction and development of new generation prostheses require the evaluation of the efficacy of LLPDs. To date, guidelines are available for testing exoskeletons [20], however, a standardized experimental method for the evaluation of LLPDs is lacking. This definitely complicates the comparison of research outcomes. Our systematic review provides an overview of currently used study designs and experimental protocols (tests and outcome measures) (Table II, Fig. 2), which lead to bottom-up (literature based) guidelines and recommendations for the evaluation of a LLPD (Fig. 3).

A familiarization trial should precede experimental trials since it reduces learning effects and enables participants to get accustomed to the novel prosthesis, the tests and measurement devices [30]. Remarkably, not even half of the papers (only 38%) included a familiarization trial. Afterwards, experimental trials include the current prosthesis, as a baseline measure, and the novel prosthesis. At lower technology readiness levels (< 6) case reports are relevant, whereas randomized controlled trials in counterbalanced order are preferred in a later stage (≥ 6 , i.e., ‘model or prototype demonstration in a relevant environment’ [140]). The comparator used in the evaluation process are participants’ current prosthesis, which are usually passive devices, but could possibly also be microprocessor-controlled or active prostheses. Therefore, if the study group is heterogenous in type-specific prosthesis, it is recommended to clearly define in- and exclusion criteria of the study group.

Walking is the most common daily activity [141], hence a key task in the evaluation process. The hallway six-minute walk test (6MWT) [142] is most frequently used because of the high test-retest reliability (Intraclass Correlation

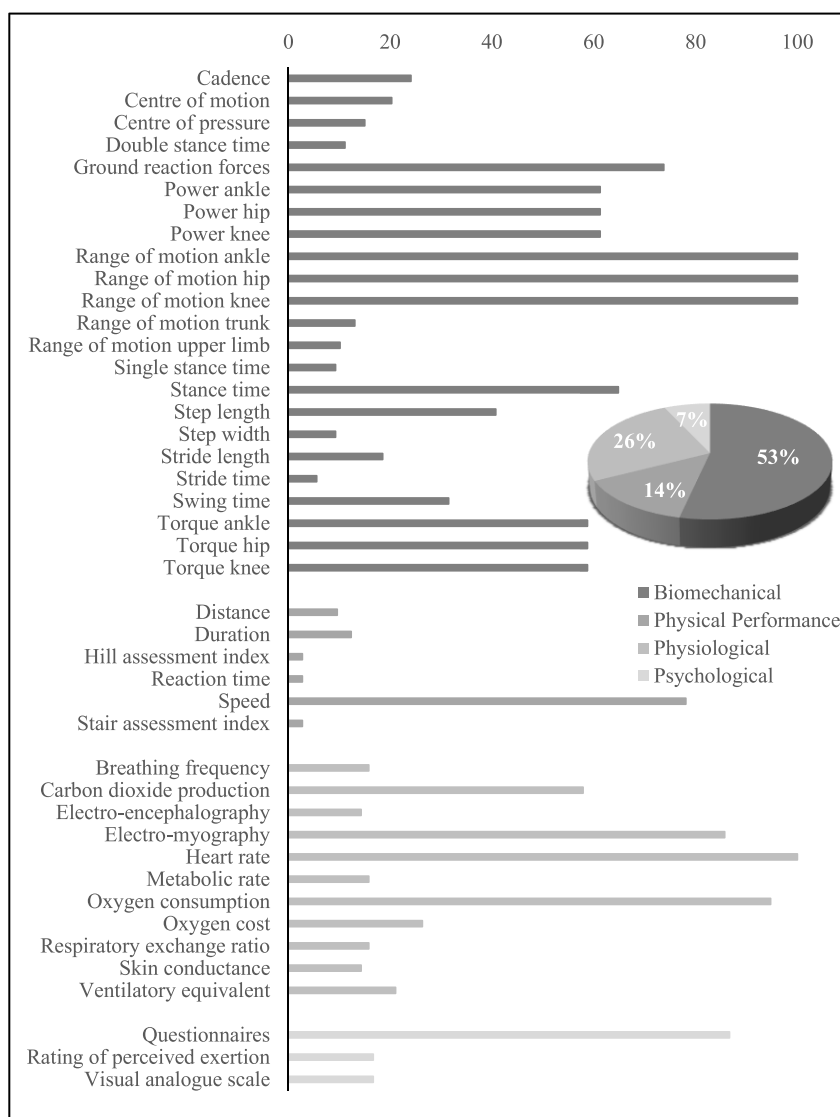


Fig. 2. Frequency analysis of relevant outcome measures for the evaluation of a lower-limb prosthetic device. The pie chart indicates the frequency of outcome measures per category.

Coefficient (ICC): 0.90-0.96 [143]). The test is performed in a hallway at a self-selected walking speed, which closely relates to the most efficient walking speed [144]. Currently, a user-driven treadmill is available to control and experience natural fluctuations in walking speed [145]. The treadmill 6MWT also showed a high test-retest reliability (ICC: 0.94-0.97 [143]) and could therefore be applied in a laboratory setting. An alternative for the 6MWT is the two-minute walk test as it has been shown to be reliable (ICC: 0.83 [146]) and a good predictor of the 6MWT [147]. The two-minute walk test is recommended during early stage prototype development or when participants are classified with a K-level < 2, and thus not capable to comfortably walk for six minutes.

Cognitive resources are used during walking and attentional demand might alter when walking with a novel device [148]. Dual task paradigms allow to investigate the attentional demand and require the participant to simultaneously conduct two tasks, from which one is cognitive and the other

motoric (e.g., walking). Research using LLPUs including dual task paradigms is scarce and includes only backward counting as cognitive task, however, other psychological studies, not included in this review, frequently applied the Stroop test [149] and the psychomotor vigilance task [150] in laboratory conditions. Thus, literature recommends to include one of the latter tests since these are easy to implement and highly reliable (ICC: 0.84-0.91 [151] and 0.84 [150], respectively).

To improve the ecological validity of the study, other daily activities should be included in the experimental protocol, such as stair climbing (ICC ascent: 1.00; descent: 0.89 [49]), slope walking (ICC uphill: 0.87; ICC downhill: 0.73 [49]) and timed up & go (ICC: 0.88 [146]). Current papers already included these tests, though, in a non-standardized manner with different modalities (e.g., inclination of the slope, number of stairs, etc.). According to literature, the following recommendations are formulated. The slope walking test [152] assesses the participants' ability to ascend and descend a 6 m long

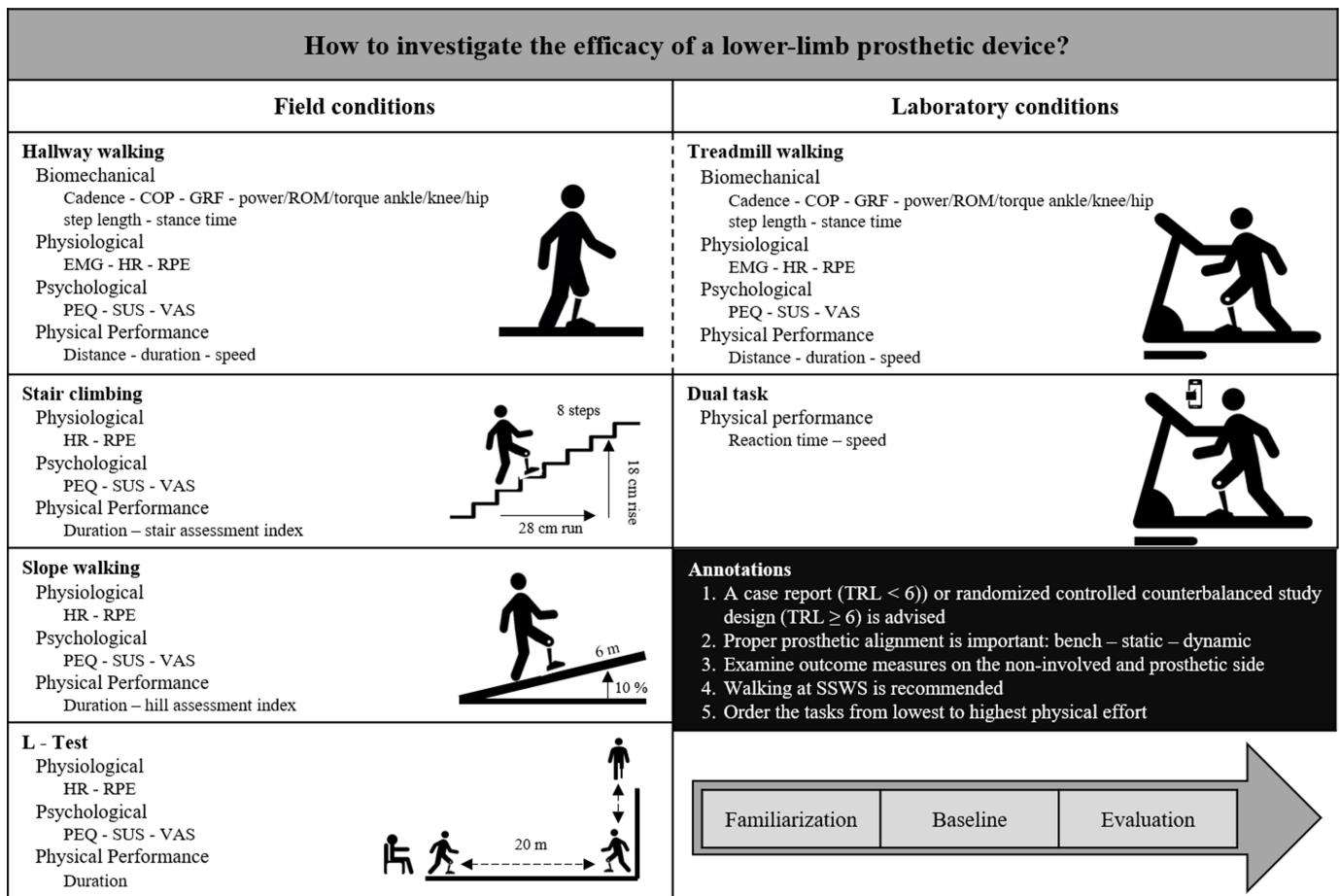


Fig. 3. Clinical practice bottom-up guidelines and recommendations to investigate the efficacy of lower-limb prosthetic devices. The black box includes points of attention regarding the experimental design (no.1), the alignment (no.2) and the procedure (no.3-5). Abbreviations: **COP** centre of pressure, **EMG** electromyography, **GRF** ground reaction forces, **HR** heart rate, **PEQ** prosthesis evaluation questionnaire, **ROM** range of motion, **RPE** rating of perceived exertion, **SSWS** self-selected walking speed, **SUS** system usability scale, **TRL** technology readiness level, **VAS** visual analogue scale for comfort and fatigue.

ramp of 10% inclination as fast as possible. For safety reasons bilateral handrails are warranted. Participants start the test in an upright position in front of the ramp and are asked to ascend the ramp, turn around on the platform, descend the ramp and return to the starting position. The stair climbing test [142] assesses the participants' ability to ascend and descend a staircase (180 cm total height, 8-step staircase with an average rise of 28 cm and run of 18 cm). Participants start in front of the staircase and are asked to ascend and descend as fast as possible. Again, bilateral handrails are required to allow support when needed. The ascending phase is initiated with the prosthetic side, while the first step of the descending phase is performed with the non-involved leg. The timed up & go [142] examines the capacity of mobility, balance and taking turns during a functional movement exercise. Participants stand up from their chair (44 cm total height, preferable without arm support and with a straight back), walk three meters, turn around, return to their chair and end in a seated position [152]. An alternative test is available, i.e., the L-test (ICC: 0.97 [153]), which requires higher physical activity of the LLPU. During the L-test, participants are asked to rise from a chair, walk through an office door, turn 90 degrees, walk down the hallway, turn 180 degrees and then return in the

same way to the seated position. A total distance of 20 meter is covered.

Primary physical performance measures are walking speed and distance covered during the 6MWT, reaction time to a stimulus during the dual task walking trial and the duration of the other physical tasks. Since fatigue influences exercise performance, the order of different tasks should be considered [154]. Currently, no data is available regarding the order of physical tests related to increasing physical effort. Therefore, it is advised to order the tests from lowest to highest physical effort taking into account heart rate and rating of perceived exertion, which are primary indicators of the intensity of effort [155]. Depending on the functionality of the novel device, some tests have priority.

Biomechanical data gathering is crucial in the evaluation process of LLPDs. This is in line with Sagawa *et al.* [22], who observed that 75% of the outcome measures were biomechanical, 21% physiological and 4% psychological. Most frequently used (>25% of the articles), and thus primary outcome measures are cadence, centre of pressure, ground reaction forces, joint torque and power of the ankle, knee and hip, step length, stance time and range of motion of the ankle, knee and hip. Force platform data collection during

walking enables researchers to assess locomotor adaptation strategies by means of spatiotemporal and kinetic measures. Kinematic measures are obtained with high speed motion capture systems (the marker based method is considered as the golden standard [156]) to analyse the human movement and calculate the angular momentum [157]. Nowadays, inertial measurement units are also applied for gait assessment in a more dynamic environment [158]. According to literature, the centre of mass and motion, double and single stance time, range of motion of the trunk and upper-limb, step width, stride length and swing time are less frequently (<25%) investigated, and thus can be considered as secondary outcome measures.

As previously mentioned, primary physiological outcome measures are heart rate and rating of perceived exertion to indicate the intensity of physical effort. They are both highly correlated [155] and easy to monitor [155]. Another indication of physical effort is EMG. Amplitudes of EMG frequency waves and power shifts of frequency bands represent muscle activation and peripheral fatigue [41], respectively. SENIAM guidelines are commonly used for sensor location, orientation and the starting posture of the clinical test [159]. A coactivation of quadriceps and hamstrings is observed during the stance phase of the gait cycle in people without an amputation [160]. In LLPUs however, and in particular individuals with a transfemoral amputation, these muscles are less innervated because of the surgical myodesis [161]. On the other hand, the adductor magnus is more active to align the hip joint and to prevent hip abduction [162]. Therefore, primary EMG outcome measures of the non-involved leg and prosthetic side are the rectus femoris, biceps femoris and adductor magnus. The most common outcome measure in the evaluation of an LLPD is oxygen consumption. However, this should be considered as secondary, since it has been shown that the accuracy of ergospirometric devices ranges between 3-10% [163] and minor differences (e.g., 4.6% decrease [128]) between conditions might be due to calibration errors, daily variability and different prosthetic fitting and alignment. Thus, small differences in oxygen consumption and carbon dioxide production might fall within the accuracy range of the ergospirometric device [128].

Subjective data gathering is an integral part of the evaluation process. Nowadays, these measures receive more attention, meaning that 28% of the gathered data is subjective in the current review compared to 7% in the review of Sagawa *et al.* [22]. Primary outcome measures are the hill [164] and stair [165] assessment index, the visual analogue scale for fatigue and comfort, and questionnaires. The latter concerns evaluating the subjective preferences and perceptions of the LLPUs with regard to the novel LLPD and should focus on four important themes, reported by Legro *et al.* [166]: (i) the fit of the socket; (ii) the mechanical functions of the prosthesis; (iii) other qualities of the prosthesis; (iv) adaptation with support from others. The aforementioned themes could be assessed with a preference ranking table, the prosthesis evaluation questionnaire [119], system usability scale [167] and visual analogue scale for comfort and fatigue [168].

Developing active prostheses are inherently more challenging than passive and quasi-passive devices. Their design is more complex and several other, non-clinical, parameters are also important. Human motion intent recognition [169] is an essential part for volitional control, especially during different terrain ambulation modes [76]. There are multiple approaches for several ambulation modes, e.g., transition time from one finite state to another [138]. Typical state-based controllers are integrated in the body of the prosthesis to detect parameters (e.g., impedance, torque or position control) and to determine the finite state [99]. In contrast to state-based controllers, continuous proportional control behaves more natural and enhance a smoother transition from stance to swing phase [138]. Thus, regarding the non-clinical evaluation of powered prostheses, human motion intent recognition should not be neglected.

Proper prosthetic fitting is a frequently reported concern of patients and of major importance since misalignment alters movement patterns and negatively affects physical performance [10]. Prosthetic alignment by the conventional method is based on three clinical steps in accordance to the manufactures guidelines: (i) bench alignment by placing the prosthesis in the middle range; (ii) static alignment to modify the socket height and orientation; (iii) dynamic alignment during gait observations and other activities to fine-tune [170]. Nowadays, technological advancements enables the certified prosthetist to align the prosthesis based on novel sensor and computer technology such as a smart-pyramid [171].

The aforementioned guidelines and recommendations are of interest for the academic and clinical world. A limitation of the current review is that guidelines are formulated for subjects aged above 18 years and with a K-level ≥ 2 [27]. Our review is based on methodological aspects of published studies and should be re-evaluated in the future.

V. CONCLUSION

Due to a lack of uniformity in experimental methodologies to evaluate a LLPD, bottom-up guidelines and recommendations are formulated. The main recommendation is to combine walking trials with other functional tests in a case report (technology readiness level < 6) and in a later stage in a randomized controlled counterbalanced study design. Literature-based primary and secondary biomechanical, physical performance, physiological and psychological outcome measures are formulated.

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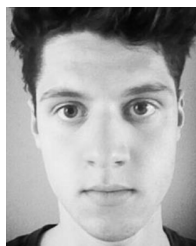
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